

Sean Lee

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EDUCATION

Keck School of Medicine, University of Southern California

Los Angeles, CA

Doctor of Medicine

Expected Graduation May 2027

University of Southern California

Bachelor of Science, Human Biology. Healthcare Studies Minor

2019 - 2022

PUBLICATIONS

Cullaro G, Lai JC, Liu T, Gomez ME, Simonetto DA, Lee BP, **Lee S**, Allegretti AS, Mazumder NR, Patidar KR, Verna EC. Machine learning–derived clinical phenotypes of ascites in cirrhosis: A multi-center latent class analysis with external validation. *Journal of Hepatology*. 2026. (*In revision for resubmission*)

Elshafei M, Pouwels S, Putnam L, Munoz X, Eini JI, **Lee S**, Lipham JC, Houghton C. Implementation of a standardized protocol for EndoFLIP utilization alters hiatus closure intraoperatively: A proof-of-concept study. 2026. (*In preparation*)

RESEARCH

USC Keck School of Medicine

Los Angeles, CA

Department of Pulmonary & Critical Care | Dr. Keith Killu

Nov 2025 - Present

Project Lead | Research Assistant

- Conceived and designed an independent retrospective study examining clinical and laboratory predictors of imaging escalation from chest X-ray to CT in hospitalized pneumonia patients, with the goal of developing evidence-based criteria to guide resource-appropriate workup.
- Formulated study hypothesis, defined patient cohort inclusion/exclusion criteria, selected candidate clinical variables, and prepared IRB submission materials.
- Leading all phases of the project including data collection, statistical analysis, and manuscript preparation for peer-reviewed publication.
- Conducted comprehensive systematic literature review and study screening for a meta-analysis comparing diagnostic accuracy of point-of-care ultrasound versus CT for pneumonia, identifying and evaluating studies across multiple databases.
- Extracted and synthesized quantitative data across included studies, assessing methodological quality and pooling effect estimates for analysis.
- Contributing to manuscript drafting and preparation for peer-reviewed submission.

Department of Transplant Hepatology | Dr. Brian Lee, Dr. Giuseppe Cullaro

Research Assistant

Sept 2025 - Present

- Performed detailed chart review and longitudinal data extraction for 150+ liver transplant patients spanning pre-listing through post-transplant outcomes, requiring individualized manual review of complex transplant records.
- Contributed to a study evaluating Cystatin C-based MELD scoring as a refinement to standard prognostic indices for improving post-transplant outcome prediction.
- Performed chart review and data extraction for a multi-center study identifying clinical predictors of cirrhosis and ascites phenotypes, contributing to a dataset spanning multiple institutions.
- Contributed to manuscript development and review examining phenotype-based risk stratification in cirrhosis; manuscript currently under revision for peer-reviewed publication.

Department of General Surgery (GLS/MIS) | Dr. John Lipham

Research Assistant

Jan 2026 - Present

- Conducting retrospective chart review and data collection for a study evaluating the impact of intraoperative platelet-rich plasma (PRP) injection on postoperative outcomes following hiatal hernia repair.
- Performed retrospective chart review and data collection evaluating intraoperative EndoFLIP utilization during antireflux surgery and its impact on real-time surgical decision-making and adjustment.
- Prepared results section and contributed to full manuscript development examining whether EndoFLIP-guided intraoperative measurements reduce postoperative reflux and dysphagia outcomes.

Department of Emergency Medicine | Dr. Alexander Sapin

Research Assistant

April 2026 - Present

- Contributing to a quality improvement initiative aimed at developing an automated flagging system to identify ED patients at risk for prolonged length of stay (LOS >21 days) to facilitate early disposition planning.
- Performing data collection and analysis of admission diagnostic codes for patients meeting LOS threshold, forming the evidential basis for future flag criteria development.

USC Alfred E. Mann School of Pharmacy | Dr. Yong (Tiger) Zhang*Research Assistant***Los Angeles, CA**

June 2020 - June 2022

- Conducted weekly literature reviews of 200+ papers and performed wet lab techniques including protein/DNA purification and cell culture.

PERTINENT EMPLOYMENT HISTORY**AHMC: Garfield Medical Center****Monterey Park, CA***Patient Liaison | Project Manager*

May 2022 - May 2023

- Partnered with physicians and nursing staff to identify and address urgent patient needs, directly improving quality of care and patient outcomes.
- Bridged communication gaps between patients, families, and clinical teams by providing real-time medical interpretation in Cantonese, Mandarin, and English.
- Conducted literature reviews to support bi-weekly COVID-19 grand rounds presentations delivered to medical and administrative staff.
- Analyzed and synthesized HCAHPS, Leapfrog, and ED-Throughput performance metrics across AHMC's 9-hospital system to inform operational decision-making.
- Prepared formal meeting minutes for senior administrative and clinical leadership.

LANGUAGES

English – Native

Chinese Cantonese – Conversational proficiency

Chinese Mandarin – Intermediate (able to take patient history and perform physical exam)

Spanish – Intermediate (able to take patient history and perform physical exam)

LEADERSHIP/VOLUNTEER EXPERIENCE**USC Keck School of Medicine****Los Angeles, CA***Office of Student Affairs | Student Affairs Representative*

Aug 2023 - Aug 2025

- Served as a primary liaison between the student body and institutional administration, advocating for student needs and driving improvements in campus resources and support services.
- Partnered with administrators to identify, propose, and implement targeted solutions addressing student well-being, including housing equity, facility enhancements, and mental health resource accessibility.
- Cultivated an inclusive environment that encouraged student engagement and empowered peers to contribute meaningfully to institutional decision-making and curricular development.

Gastroenterology Student Interest Group | Vice President

Aug 2023 - Dec 2024

- Spearheaded programming initiatives by coordinating with faculty and clinical speakers to curate lunch talk series, delivering career-relevant insights to students pursuing medicine.
- Led targeted outreach to identify and compile a diverse roster of research-active faculty, connecting students with mentorship and research opportunities across a broad range of subspecialties and interests.

USC Photography Club | President | Events Director

Aug 2020 – Aug 2023

- Led executive team of 15 people to organize events and excursions for 20+ participants, promoting community engagement, Managed operations across event planning, marketing, and logistics departments.
- Coordinated large-scale events for 2,000+ members, including multi-day retreats and workshops.

Dmanisi Archaeological Field School**Los Angeles, CA***Field Researcher*

Jun 2019 - Aug 2019

- Participated in active excavation at the Dmanisi site, one of the most significant *Homo erectus* fossil sites outside Africa, gaining hands-on exposure to paleoanthropological methods and human evolutionary history.
- Attended lectures on hominin morphology, species differentiation, and the anthropological significance of early human migration.
- Observed firsthand the burden of undertreated disease in rural Georgia — including cases of polio and other conditions rarely encountered in the U.S. — deepening my commitment to preventive medicine and global health equity.

Alhambra Hospital**Alhambra, CA***Emergency Medicine Shadowing | Volunteer*

July 2020 – Dec 2022

- Observed clinical decision-making in high-acuity emergency settings across diverse patient populations, and assisted nursing staff with patient care, vitals, and unit operations.

Connect Tech Network**San Francisco, CA***Tutor Volunteer*

June 2020 – June 2021

- Taught digital literacy (Zoom, Google tools) in Cantonese to elderly clients during COVID-19 to improve accessibility of services.

PROJECTS

AI-Powered Clinical Documentation Tool (Independent Project)

April 2026

- Developed a HIPAA-compliant note-taking application to generate standardized physician documentation using keyword recognition and dictation.
- Designed to improve efficiency and consistency in clinical note-writing without storing patient data.

Antibiotic Stewardship Tool (Independent Project)

May 2026

- Developed an interactive web-based reference tool to support evidence-based antibiotic prescribing in Internal Medicine. Designed for rapid bedside use, delivering setting-appropriate regimens in 3 clicks, across 30+ infection types and 9 organ systems (outpatient, floor, ICU). Features dynamic adjustments for penicillin allergy and MRSA risk factors, with stewardship guidance on de-escalation, culture-directed therapy, and treatment duration per current IDSA/ASHP guidelines. Deployed as a publicly accessible resource for medical learners. [<https://github.com/infer321/AntibioticStewardshipTool>]

Internal Medicine Clinical Scoring Calculator (Independent Project)

May 2026

- Developed a web-based calculator integrating 15 validated clinical scoring tools used in internal medicine, including HEART, CURB-65, MELD-Na, NIH Stroke Scale, Wells PE/DVT, SOFA, APACHE II, GRACE, and CHA₂DS₂-VASc. Organised by specialty with real-time score interpretation and evidence-based clinical guidance. Built in HTML/CSS/JavaScript; mobile-optimised and deployable offline. [<https://github.com/infer321/InternalMedCalc>]

CERTIFICATES

Quality Improvement Certificate | USC Keck School of Medicine

2026

- Completed formal training in QI methodology, including process improvement frameworks and patient safety principles applicable to clinical practice.

Certificate of Merit Piano, Level 10 (Highest Level) | Music Teachers' Association of California

2018

- Achieved the highest level of the statewide classical piano assessment, demonstrating years of disciplined technical and musical development

PROFESSIONAL INTERESTS

Piano, Weightlifting, Cycling, Running, Photography.